

$X \backslash Y$	y_1	y_2	y_3
x_1	a	b	c
x_2	d	e	f
x_3	g	h	i

$a+b+c$
 $d+e+f$
 $g+h+i$
 $a+d+g$ $b+e+h$ $c+f+i$

$$\begin{cases} a, b, c, d, e, f, g, h, i \geq 0 & a, b, c, d, e, f, g, h, i \leq 1 \\ a+b+c+d+e+f+g+h+i = 1 \end{cases}$$

$(a+b+c)(a+d+g)$	$(a+b+c)(b+e+h)$	$(a+b+c)(c+f+i)$
$(d+e+f)(a+d+g)$	$(d+e+f)(b+e+h)$	$(d+e+f)(c+f+i)$
$(g+h+i)(a+d+g)$	$(g+h+i)(b+e+h)$	$(g+h+i)(c+f+i)$

$$P[X=x_1] = \sum_{j=1}^3 P[X=x_1 | Y=y_j] P[Y=y_j] = \sum_{j=1}^3 P[X=x_1, Y=y_j]$$

$$P[X=x_i] = \sum_{j=1}^3 P[X=x_i, Y=y_j] \rightarrow \text{le marginali di } X \text{ si ottiene sommando per righe}$$

$$P[Y=y_j] = \sum_{i=1}^3 P[X=x_i, Y=y_j] \rightarrow \text{le marginali di } Y \text{ si ottiene sommando per colonne}$$

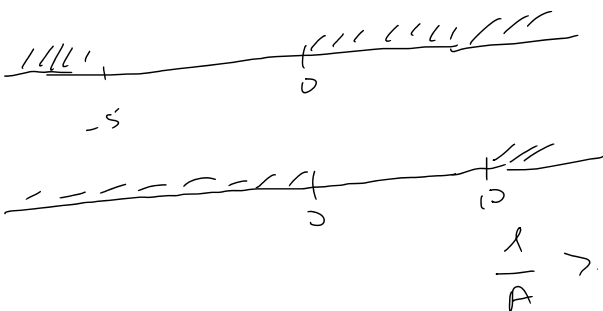
$$a \log_2 \frac{a}{(a+b+c)(a+d+g)} + b \log_2 \frac{b}{(a+b+c)(b+e+h)} + c \log_2 \frac{c}{(a+b+c)(c+f+i)} + \dots$$

si attende
 "ineguaglianze"
 per cui
 $a, b, c, d, e, f, g, h, i$
 fossero nulli

$$0.2 + \frac{1}{A} \quad 0.1 - \frac{1}{A}$$

$$\begin{cases} 1 \geq 0.2 + \frac{1}{A} > 0 \\ 1 \geq 0.1 - \frac{1}{A} > 0 \end{cases} \Rightarrow \begin{cases} \frac{1}{A} > -0.2 \\ \frac{1}{A} < 0.8 \end{cases} \Rightarrow \begin{cases} A > 0 \\ -5 < A \end{cases}$$

$$\frac{1}{A} > -0.9 \Rightarrow \begin{cases} A < 0 \\ \frac{10}{A} < 1 \\ A > 10 \end{cases}$$



$$\frac{5}{A} > -1 \Leftrightarrow \begin{cases} A > 0 \\ A < 0 \dots 5 < -A \Leftrightarrow A < -5 \end{cases}$$



$$\begin{bmatrix} P^{(1)} \\ P_1 \\ P^{(i)} \end{bmatrix}$$

$$[\lambda I]_i = \begin{bmatrix} 0 \\ 0 \\ \lambda \\ 0 \end{bmatrix}$$

$$[P^T]_i$$

$$\begin{bmatrix} P_1^{(i)} \\ P_2^{(i)} \\ \vdots \\ P_n^{(i)} \end{bmatrix}$$

$$P_i^{(i)}$$

$$[\lambda I - P^T]_i = \begin{bmatrix} -P_1^{(i)} \\ -P_2^{(i)} \\ \lambda - P_i^{(i)} \\ \vdots \\ -P_n^{(i)} \end{bmatrix}$$

$$\begin{aligned} & -P_1^{(i)} - P_2^{(i)} - \dots - P_n^{(i)} \\ & -P_n^{(i)} = \\ & = \lambda - (P_1^{(i)} + P_2^{(i)} + \dots + P_n^{(i)}) \\ & = \lambda - 1 \end{aligned}$$

$$P^T \underline{v}_i = \lambda_i \underline{v}_i$$

$$\lambda_i = 1$$

$$P^T \underline{v}_i = \underline{v}_i$$

$$\underline{v}_i = \begin{bmatrix} v_{i,1} \\ v_{i,2} \\ \vdots \\ v_{i,n} \end{bmatrix}$$

$$2 \quad \underline{w}_i = \frac{1}{2} \underline{v}_i$$

$$(\underline{w}_i)_j = \sum_{j=1}^n v_{i,j}$$

$$P^T \underline{w}_i = P^T \left(\frac{1}{2} \underline{v}_i \right) = \frac{1}{2} P^T \underline{v}_i = \frac{1}{2} \underline{v}_i = \underline{w}_i$$

$$w_{i,j} = \frac{1}{2} v_{i,j} \quad \sum_{j=1}^n w_{i,j} = \sum_{j=1}^n \frac{1}{2} v_{i,j} = \frac{1}{2} \sum_{j=1}^n v_{i,j} = \frac{1}{2} = 1$$

$$\underline{f}[n+2] = (P^T)^T \underline{f}[n] \quad (P^T)^T = (P^T)^T$$

$$P[X(n+2)=i] = \sum_{j=1}^n P[X(n+2)=i | X(n)=j] P[X(n)=j]$$

$$\begin{bmatrix} P[X(n+2)=1] \\ P[X(n+2)=2] \\ \vdots \end{bmatrix} = \begin{bmatrix} P[X(n+2)=1 | X(n)=1] & P[X(n+2)=1 | X(n)=2] & \dots \\ P[X(n+2)=2 | X(n)=1] & P[X(n+2)=2 | X(n)=2] & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} P[X(n)=1] \\ P[X(n)=2] \\ \vdots \end{bmatrix}$$

$$\begin{bmatrix} P[X(n+1)=i] \\ \vdots \\ P[X(n+1)=r] \end{bmatrix} = \underbrace{\begin{bmatrix} P[X(n+2)=i|X(n)=1] & P[X(n+2)=i|X(n)=2] & \dots & P[X(n+2)=i|X(n)=r] \\ \vdots & \vdots & \ddots & \vdots \\ P[X(n+2)=r|X(n)=1] & P[X(n+2)=r|X(n)=2] & \dots & P[X(n+2)=r|X(n)=r] \end{bmatrix}}_P \begin{bmatrix} P[X(n)=1] \\ \vdots \\ P[X(n)=r] \end{bmatrix}$$

$$\underline{f}(n+2) = \begin{pmatrix} P^T \end{pmatrix} \underline{f}(n+1) = \begin{pmatrix} P^T \end{pmatrix} \left(\begin{pmatrix} P^T \end{pmatrix} \underline{f}(n) \right) = \left(\begin{pmatrix} P^T \end{pmatrix}^2 \right) \underline{f}(n) = \left(\begin{pmatrix} P^2 \end{pmatrix}^T \right) \underline{f}(n)$$

$$\underline{f}(n+2) = \begin{pmatrix} P^2 \end{pmatrix}^T \underline{f}(n)$$

$$\underline{f}(n+k) = \begin{pmatrix} P^T \end{pmatrix} \underline{f}(n+k-1) = \begin{pmatrix} P^T \end{pmatrix} \left[\begin{pmatrix} P^T \end{pmatrix} \underline{f}(n+k-2) \right] = \left(\begin{pmatrix} P^2 \end{pmatrix}^T \right) \underline{f}(n+k-2) = \left(\begin{pmatrix} P^2 \end{pmatrix}^T \right) \left[\begin{pmatrix} P^T \end{pmatrix} \underline{f}(n+k-3) \right] = \left(\begin{pmatrix} P^3 \end{pmatrix}^T \right) \underline{f}(n+k-3)$$

$$A \underline{v}_i = \lambda_i \underline{v}_i$$

$$\begin{aligned} \underline{A}^2 \underline{v}_i &= \begin{pmatrix} A & A \end{pmatrix} \underline{v}_i = A (A \underline{v}_i) = A (\lambda_i \underline{v}_i) = \lambda_i A \underline{v}_i = \lambda_i (\lambda_i \underline{v}_i) = \lambda_i^2 \underline{v}_i \\ \underline{A}^k \underline{v}_i &= \lambda_i^k \underline{v}_i \end{aligned}$$

$$\underline{A}^n \underline{v}_i = \underline{A}^{n-1} (A \underline{v}_i) = \lambda_i \underline{A}^{n-1} \underline{v}_i = \lambda_i (\lambda_i^{n-1} \underline{v}_i) = \lambda_i^n \underline{v}_i$$

$$\underline{A} = \sum_{i=1}^n \lambda_i \underline{v}_i \underline{v}_i^T$$

$$\boxed{\underline{A} \underline{u} = \underline{u}}$$

$$\underline{A}^k \rightarrow \gamma \underline{u} \underline{u}^T$$

$$\underline{A}^k = \sum_{i=1}^n \lambda_i^k \underline{v}_i \underline{v}_i^T$$

$$\begin{pmatrix} P^T \end{pmatrix} \underline{f} = \underline{f}$$

$$\left(\gamma \underline{u} \underline{u}^T \right) \underline{f} = \underline{f} \Rightarrow \underline{f} \propto \underline{u}$$

$$\gamma \underline{u} (\underline{u}^T \underline{f}) = \underline{f}$$

$$\gamma \underbrace{[\underline{u}^T \underline{f}]}_{\gamma'} \underline{u} = \underline{f}$$

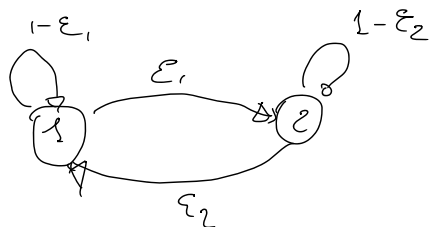
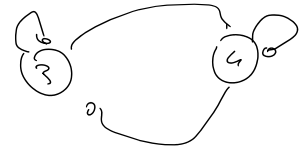
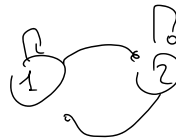
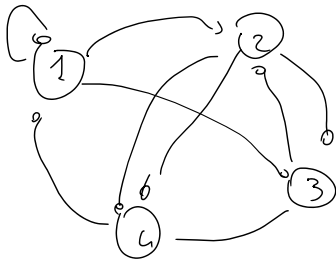
$$\lim_{k \rightarrow \infty} \underline{f}(n+k) = \lim_{k \rightarrow \infty} \left\{ \left(\begin{pmatrix} P^k \end{pmatrix}^T \right) \underline{f}(n) \right\} = \gamma \underline{u} [\underline{u}^T \underline{f}(n)] = \gamma' \underline{u}$$

$$\lim_{k \rightarrow \infty} \frac{1}{k} \sum_{n=0}^{k-1} P^n(x) = \int \dots$$

$$= \left\{ \lim_{k \rightarrow \infty} P^n \right\}^+ \underline{f}(n) = \left[\gamma \underline{u} \underline{u}^T \right] \underline{f}(n) = \gamma \underline{u} \left[\underline{u}^T \underline{f}(n) \right] =$$

$$\left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right] \left[\begin{array}{c} \xrightarrow{\underline{u}^T} \\ \vdots \\ \vdots \\ \vdots \end{array} \right] \left[\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right] = \gamma \left[\underline{u}^T \underline{f}(n) \right] \underline{u} = \gamma' \underline{u}$$

$$\gamma' = \gamma \left[\underline{u}^T \underline{f}(n) \right]$$



$$P = \begin{bmatrix} 1-\epsilon_1 & \epsilon_1 \\ \epsilon_2 & 1-\epsilon_2 \end{bmatrix}$$

$$P^T = \begin{bmatrix} 1-\epsilon_1 & \epsilon_2 \\ \epsilon_1 & 1-\epsilon_2 \end{bmatrix}$$

$$\begin{bmatrix} 1-\epsilon_1 & \epsilon_2 \\ \epsilon_1 & 1-\epsilon_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix}$$

$$\begin{cases} (1-\epsilon_1)v_1 + \epsilon_2 v_2 = \lambda v_1 \\ \epsilon_1 v_1 + (1-\epsilon_2)v_2 = \lambda v_2 \end{cases}$$

$$\begin{bmatrix} 1-\epsilon_1-\lambda & \epsilon_2 \\ \epsilon_1 & 1-\epsilon_2-\lambda \end{bmatrix} \underline{v} = \underline{0}$$

$$(1-\epsilon_1-\lambda)(1-\epsilon_2-\lambda) - \epsilon_1 \epsilon_2 = 0$$

$$\lambda^2 - [(1-\epsilon_1) + (1-\epsilon_2)]\lambda + (1-\epsilon_1)(1-\epsilon_2) - \epsilon_1 \epsilon_2 = 0$$

$$\lambda_1, \lambda_2$$